ClassLens

Turning Mock-Exam Data into Personalised Student Study Plans a Cram School (or Tutor) Can Sell

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Big Data Systems — Final Project (Spring 2026, National Taiwan University)

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GitHub repository: https://github.com/petercechung/classlens

Live demo: https://classlens.cechung.com

Demo walkthrough: teacher dashboard (`/`), student portal (`/student.html`), and example report (`/report?student=S1004`).

Executive summary

Taiwanese cram schools (buxiban) run weekly mock exams and then discard the answer-level data after computing a raw score. **ClassLens** ingests that existing answer-sheet data and turns it into a **personalised study plan for each student**: the weak topics that are also heavily tested, ranked by payoff, with a two-week practice sequence and an **estimated attainable score lift** if the plan is followed — delivered as a print-ready report. Internal-efficiency framing (saving teacher time) is too weak a motivator on its own; the sellable value is the personalised recommendation, which a school uses to **recruit and retain** (showing parents exactly what their child should work on and the expected gain). The same per-student report works with no school at all, so an **independent tutor** can use it directly — which also seeds the first real data and demand evidence. The system does the routine diagnosis, recommendation and report-writing; the teacher keeps the teaching.

1. Target customer

Primary customer: owners / head teachers of **independent and small-chain, exam-focused cram schools serving junior- and senior-high students** — the segment that runs frequent mock exams and lives or dies on results and parent trust. **Beachhead / design partner: independent tutors** (including the author's own tutoring students), who need no procurement decision and let the product prove value on day one.

Why personalisation, not admin efficiency, is the wedge. Saving a teacher an hour is a weak purchase trigger — teacher time is cheap and the pain is mild. What a school will pay for is a **per-student personalised plan with an explainable lift estimate**, because that is something it can put in front of parents to recruit and to justify staying enrolled. A peer-tutor interview reinforced this wedge: some students had left large cram schools for private tutoring because standardised large-class resources were not customised enough. The deliverable is therefore a sales/retention asset, not an internal report.

Why this wedge. Taiwan has roughly **12,500 academic (wenli) cram schools (~17,500 operating entities)**, an industry estimated at **NT$150–170 billion/year**, highly fragmented into tiny operators with budget but no data/IT team — the textbook SMB SaaS buyer. The junior/senior-high exam-prep slice is where exam stakes, parent willingness-to-pay and the value of diagnostics are all highest (the elementary majority is larger in headcount but lower in analytics need).

The job today. Teachers cull wrong answers and write parent feedback by hand, or rely on a raw score. Walled content platforms (PaGamO, Yincai Net) only analyse *their own* question banks, not the school's own mock exams. ClassLens analyses the data the school already owns — the unserved gap.

Secondary / expansion: small-chain head offices (one rollout covers many branches) and a later admissions-portfolio (xuexi licheng) analytics module. Students/parents are the data source and the value delivered, *not* the direct payer.

2. Evidence of demand and willingness to pay

2.1 Data-acquisition process

Following the Homework-2 methodology, demand is established from four sources:

- **Public / market data:** Ministry-of-Interior and industry statistics on cram-school counts, sector value and per-capita tutoring spend (below).
- **Competitor pricing benchmarks:** existing tools' published prices, used as willingness-to-pay anchors.
- **A reproducible survey instrument (not yet a full field result):** a 6-question owner survey and a 10-school interview guide (in the repo, `survey/`) plus an illustrative response sample (`survey/results_sample.csv`) showing the intended analysis — minutes/exam spent on diagnosis and stated WTP. A full field round with the author's tutoring / cram-school contacts is the planned next step.
- **Practitioner workflow observation:** the author already works as an independent math tutor, so the first user observation is the author's own workflow: after a quiz, wrong answers must be grouped by topic, converted into next-practice advice and explained to students/parents (written up in the repo, `survey/interview_notes.md`). This validates the workflow pain, but on its own it is single-practitioner evidence rather than market proof — hence the survey instrument above for a wider round.

The quantitative claims below therefore rest on the *real* public-market and competitor-pricing data; the survey instrument converts that into school-level willingness-to-pay once run.

2.2 Quantified demand signals

Signal	Value	Source
Academic cram schools (2025)	~12,539	PTS
Sector value (estimate)	≈ NT$150 billion / yr	SEF
JH out-of-school / tutoring spend	≈ NT$71k / yr; tutoring ≈ NT$60k / yr	MOE via CNA
Tutoring-management SaaS pricing (WTP anchor)	NT$1,500–10,000 / mo	VibeAI / SchoolTracs

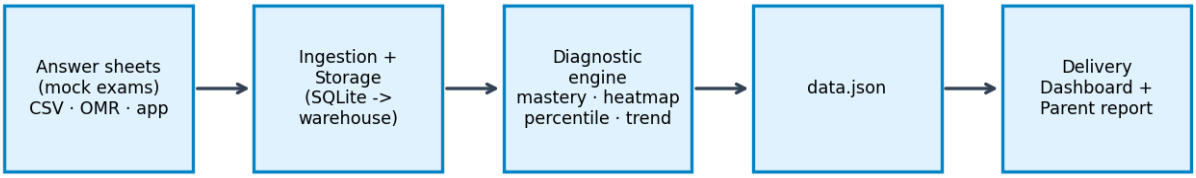
Willingness to pay (pricing model). B2B subscription priced per branch: **NT$3,000–8,000 / month / branch**, or per enrolled student per term. This sits inside an established budget line: tutoring-management systems (e.g. SchoolTracs and the systems surveyed by VibeAI) already charge schools roughly **NT$1,500–10,000 / month**, so schools are used to paying for back-office SaaS. The value is concrete: with per-student tutoring spend around NT$60,000/year, retaining a single student who would otherwise leave covers many months of subscription, and the personalised report doubles as a recruitment tool.

2.3 Why the author can test this wedge

The author's tutoring work is useful as a **beachhead pilot**: it shows the routine workflow ClassLens automates (diagnose wrong answers, identify weak topics, write student/parent-facing next steps) and provides a low-friction first dataset. The peer-tutor interview adds **access**, not statistical proof: it points to students who already left large cram schools because customised resources were weak, giving the project a realistic first interview and pilot channel. The report therefore treats this as founder-market fit and early user access, while the pricing claim remains anchored by public market size and tutoring-management SaaS benchmarks.

3. System design

ClassLens is a four-stage pipeline: **ingestion** → **storage** → **processing** → **delivery**.



3.1 Data sources & ingestion (low friction)

Adoption hinges on ingesting the data a school *already has*. Schools do not keep clean long-format tables; they have a card-reader (OMR) or Excel export: one row per student, one column per question, cells = the chosen answer (or a correct/incorrect mark), plus a one-row answer key. The importer (`import_data.py`) converts exactly that into the pipeline's inputs. Crucially, **item difficulty is computed automatically** from the empirical pass rate, so the only one-time input is a question→topic map (or an annotated answer key). This removes the main barrier to a teacher trying it on an existing exam.

No content lock-in: ClassLens analyses the school's own exams (unlike walled platforms that require their question bank).

3.2 Storage

SQLite for the demo; the schema (students, items, responses) and the GROUP-BY aggregations are warehouse-agnostic. At scale, `responses` is the only large, append-only table, partitioned by (school, term).

3.3 Processing — the diagnostic engine

Mastery is **difficulty-weighted** (an IRT-style scheme): a correct answer on a hard item counts more, so the engine recovers a student's true weak topics rather than echoing the raw score. It computes topic mastery, a class weakness heatmap, cohort percentile and a term trend. The core output is a **personalised study plan**: topics are ranked by *gap x exam-weight* (a weak topic that is also heavily tested has the highest payoff) and sequenced into a two-week plan. Each topic carries an **estimated attainable lift** — an *explainable heuristic* (raise a weak topic by a capped step toward a target, weighted by its share of the exam), presented as an **opportunity estimate, not a validated prediction**. All metrics are explainable SQL aggregations + light Python: cheap, and easy for teachers and parents to trust.

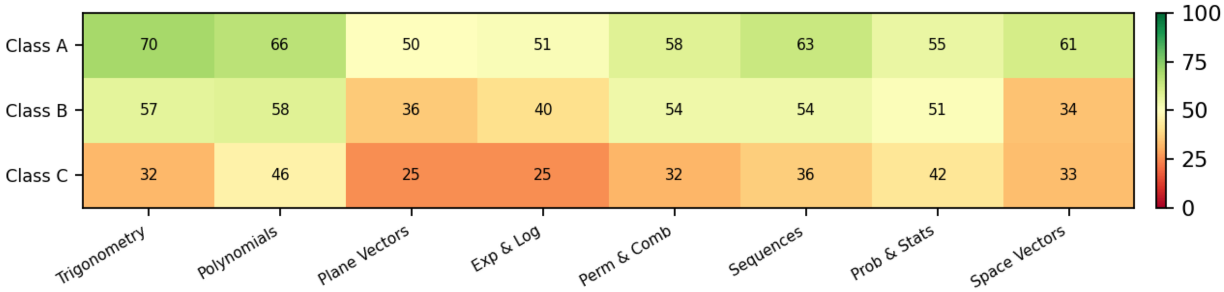


Figure: real pipeline output — class weakness heatmap across the three demo classes (mean mastery %, latest mock exam).

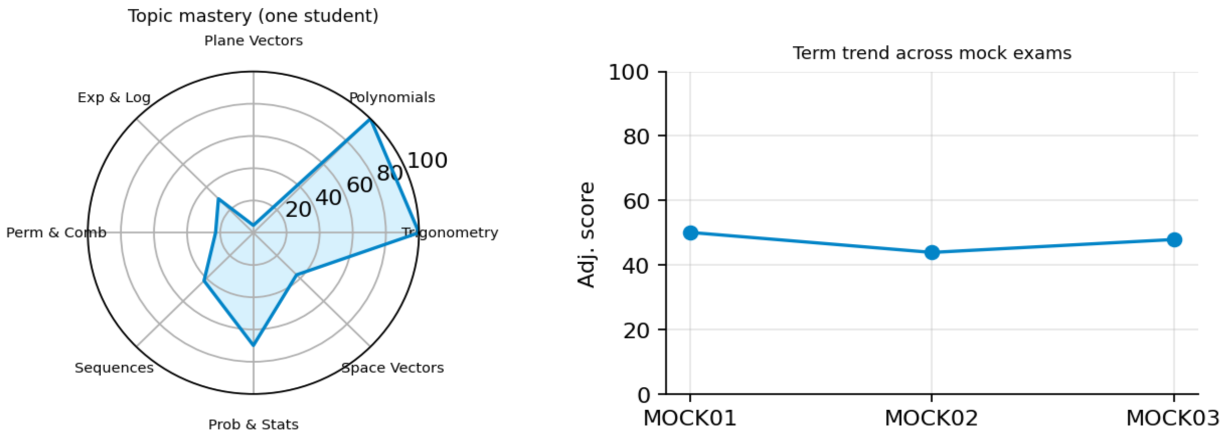


Figure: real pipeline output — one student's topic-mastery radar and term trend, as shown in the parent report.

3.4 Delivery

Three static views that read `data.json` at runtime: a **teacher dashboard** (class heatmap, ranked students, weak topics); a **student/parent portal** (`student.html`) where a student selects their identity and opens their own report — so the product is not teacher-only; and the **personalised report** itself (`report.html?student=ID`; radar, trend, two-week plan, estimated lift). In production the portal is login-gated so a student sees only their own report (PDPA). All deploy as a read-only static bundle (Vercel).

Route	What the grader can verify
`/`	Teacher dashboard with class tabs, KPI cards, weakness heatmap and ranked student list.
`/student.html`	Student/parent entry point for opening an individual report without using the teacher dashboard.
`/report?student=S1004`	Example personalised report with radar chart, trend chart and two-week study plan.

3.5 Scalability & cost

The workload is bursty and batch-shaped: each exam upload triggers one diagnostic run, embarrassingly parallel by school/exam — no always-on streaming needed. At 100x (~100 branches, ~25k students, ~30M responses/term) the relational store moves to Postgres + object storage for raw scans; reports are pre-rendered on upload.

4. Go-to-market difficulties (bonus)

The hardest part is not building the dashboard; it is making a small cram school trust, adopt and repeatedly pay for a diagnostic workflow. The table below maps the main promotion risks to concrete mitigations.

Difficulty	Why it is hard	ClassLens mitigation
Trust & adoption	Teachers and parents will not trust a black-box AI score, especially if it claims a precise grade increase.	Run on the school's own exams; show topic mastery, exam weight and the heuristic lift calculation. The teacher keeps final judgment and can add comments.
Data acquisition friction	Small cram schools have messy Excel / OMR exports and do not want to change their teaching system.	Importer accepts wide answer-sheet CSV/XLSX, computes difficulty from pass rate and requires only a question-topic map. One-class pilot can run from an existing exam.

Difficulty	Why it is hard	ClassLens mitigation
Privacy / PDPA	Student performance data is sensitive and parents may object to cross-school data resale.	Use anonymised student IDs, school-as-data-controller, single-tenant or on-prem deployment for conservative schools and no resale of student-level data.
Cold start	Without real school data, the product has little proof that reports improve retention or parent conversations.	Start with the author's tutoring workflow and one design-partner class; use before/after parent-report examples as sales proof before scaling to 10-school interviews.
Competition & moat	PaGamO / Yincai Net and management systems already exist; large chains may build internal analytics.	Avoid competing as a content platform or admin system. The wedge is analysing the school's own exams and turning that data into parent-facing retention material for independents and small chains.
Unit economics	Small branches have limited budgets, so heavy model or support costs would break the business.	Batch processing on small append-only response tables keeps marginal compute low. NT$3k-8k/month is justified if one retained student covers months of subscription.

References

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